Applied Corporate Finance

Week #5A – Tesla-SolarCity

Class-Case Discussion

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Plan of Attack

- **Background**
- **Discussion/Proposed Solution to the Case**
- **What Happened?**

Objectives

- Understand that **mergers should create value**, put simply: 2 + 2 > 4. There should be positive synergies, after accounting for transaction costs.

- **Valuing the merger and assessing market reactions.**

- **Allocation of value between the capital providers of both firms**, because SolarCity is arguably distressed, the case highlights how its capital structure impacts the overall allocation of value in the deal.

- **Corporate governance:** the case provides a rich setting to discuss corporate governance issues that arise from a strong-willed founder and owner CEO in Elon Musk, who had several interests in both companies: as a shareholder and as a bond-holder.

Background

- On June 21, 2016, Tesla Motors, Inc. offered to acquire SolarCity, bringing CEO Elon Musk one step closer to achieving his vision of moving the world from a hydrocarbon-based economy to a more sustainable solar-electric one.

- Markets and analysts were critical in their reaction to the announcement:
  - some thought the deal would be a distraction to Tesla management at a critical time;
  - others thought it was a "bailout" of SolarCity.

- Following weeks of due diligence, Tesla and SolarCity finalized their merger agreement and worked to justify the transaction on July 29.

- Joan Banister, a financial advisor, must analyze the deal and prepare to address her clients' concerns about the value of their financial positions in Tesla and SolarCity.

Tesla

- Headquartered in Palo Alto, California, Teslaa was founded in 2003 with the near-term goal of building sports cars powered by lithium-ion batteries.

- Elon Musk had been a major investor in Tesla's Series A financing round and became its Chairman in 2004. In 2008, Musk assumed the role of CEO and, as of July 2016, owned over 20% of Tesla's equity.

- as of June 30, 2016, had a market capitalization of over $31 billion, ranking it among the largest U.S. automakers based on market capitalization.

- Since the introduction of its Model S sedan in 2012, Tesla had grown quickly and its two main electric cars, the Model S sedan and the Model X sports utility vehicle, were at the top of the luxury market in terms of critical reviews and performance. In 2015, Tesla delivered 50,000 vehicles, bringing total deliveries since 2012 to 107,000.

- Tesla planned to introduce a lower-cost, “mass-market" model (the "Model 3") in 2017 and the firm anticipated the new Model 3 would be in large demand. With the introduction of the Model 3, Tesla expected to increase its annual production of new vehicles to 500,000 units by 2018.

Tesla – Gigafactory

- To support the production of its vehicles, Tesla broke ground on its “Gigafactory" in 2014.

- According to projections, the Gigafactory would make Tesla the world's largest producer of lithium- ion batteries and provide sufficient capacity to supply all of the batteries for its automobiles.

- In addition, the Gigafactory would also manufacture the Tesla “Powerwall", a rechargeable battery system designed for home energy storage. Utilizing the same lithium-ion battery technology that powered Tesla vehicles, the Powerwall took energy from solar sources or the electric grid and allowed residents to power their homes at night or in the event of an outage, all from the touch of a mobile-phone based application.

- Importantly, the Powerwall provided the link to Musk's broader vision of a "sustainable energy" future.

Elan Musk’s vision of a "sustainable energy"

- Musk had a long-term vision of providing consumers with a viable alternative to fossil fuels. In his 2006 “Secret Tesla Motors Master Plan”, Musk wrote:

  “the overarching purpose of Tesla Motors … is to help expedite the move from a mine-and-burn hydrocarbon economy toward a solar electric economy." Summarizing his plan, Musk concluded:

  So, in short, the master plan is: Build sports car

  Use that money to build an affordable car

  Use that money to build an even more affordable car

  While doing above, also provide zero emission electric power generation options”

- Musk explained that there were three important components to a sustainable energy future: generation, storage, and consumption. Tesla's Powerwall allowed consumers to efficiently store the energy they harvested from the sun, and the fleet of Tesla's electric vehicles provided a more environmentally friendly way to consume this energy.

- To this end, Musk encouraged his cousins, Peter and Lyndon Rive, to start a company devoted to solar power generation. The Rives founded SolarCity in 2006 with Musk as Chairman. SolarCity was the last piece of the puzzle in terms of providing low-emission power generation. Musk stated: “This is the integrated future. You've got an electric car, a Powerwall, and a Solar Roof. It's pretty straightforward, really."

SolarCity

- SolarCity designed and manufactured solar panels for residential and commercial use and was founded on the idea that solar panels would eventually replace utility-generated power as the predominant source of energy for households, businesses, and governments.

- Since its inception, SolarCity had installed more than 230,000 solar energy systems for customers in 27 different states.

- In 2015, SolarCity reported revenues of $400 million, up 57% from the previous year and, as of June 21, 2016 had a market capitalization of $2.2 billion.

- Since 2008, the “SolarLease” had been a key element of SolarCity's sales strategy. When a customer chose this financing option, SolarCity would install solar panels on the customer's home at no upfront cost and, in return, the customer would enter into 20-year lease agreement with SolarCity. Because SolarCity incurred the upfront installation cost but realized the lease payments over 20 years, its rapid growth had caused its cash flow from operations to be negative.
  - For example, during the first six months of 2016, SolarCity's cash flow from operations was negative $389 million. For this reason, SolarCity was highly dependent on external funding sources, specifically debt capital.

SolarCity (Cont.)

- In December 2015, an adverse regulatory change in Nevada caused SolarCity to revise its guidance for the first quarter of 2016 downward, the ruling caused analysts to question whether SolarCity would be able to achieve its 2016 projections.

- This development, combined with declining oil and gas prices that made solar relatively less attractive, led some analysts to express concern about SolarCity's ability to meet its growth forecasts and raise debt at reasonable terms. To make matters worse, in March 2016, a rumor regarding the potential bankruptcy of a major competitor caused the stock prices of several solar firms, including SolarCity, to fall significantly.
  - By June 21, 2016, SolarCity's stock price was down 52% from its value on December 31, 2015 and
  - Its convertible notes were trading at prices averaging 62% of par value, implying a yield of approximately 25% on the straight-bond component.
    - In comparison, Tesla's convertible notes implied a straight-bond yield of 7.4% and the yield on CCC rated corporate bonds was 14.4% at the time.

Motivation for the merger

Motivations for Tesla-SolarCity Merger

Pros of merger:

- Completes Grand Vision (sustainable energy future).
- Combined Retail Networks.
- Cost Savings ($150m/year ; PV= € 950m).
- Liquidity Crisis: Great Price.
- In Musk We Trust (genius?).

Cons of merger:

- Distracts from Model 3 (scaling up is big deal).
- Could do JV / Contracts.
- Savings are relatively Small
- Liquidity Crisis: Bailout.
- Musk conflicted (owns TSLA & SCTY).

Merger Rationale and Structure (Overview)

Tesla		+	SolarCity		> = <	CombinedCo / TSLA + SCTY	
Opeating Assets	Debt		Opeating Assets	Debt		Opeating Assets TSLA SCTY	Debt Old TSLA Old SCTY
	Equity			Equity		Synergies	CombinedCo Equity

Rationale for Tesla-SolarCity Merger

How Can Tesla-SolarCity Merger Create Value?

- Are there **synergies**?
  - Cost savings?
  - Revenue increasing?
  - Risk reducing?
- Is SolarCity **mis-managed**?
- Is SolarCity **under-valued**?
- Are there any **private motives** for managers or some owners?

Merger terms

Merger terms

- **Offer:** June 21, 2016 (subject to due diligence).
- **Agreement:** July 29, 2016.

- All Equity Deal.

- **Fixed Exchange Ratio:**
  - **Offer: 0.122 - 0.131** (Avg. = 0.1265) Tesla Share for each SolarCity share.
  - **Agreement: 0.110** TSLA per each SCTY.

- 45-Day "Go Shop" Period.

- Approval of both TSLA and SCTY shareholders
  - "Disinterested Shareholders" only.
  - Excludes Musk (and others).

Value of Initial Offer on June 21, 2016 (Ex-Ante)

Tesla alone				+	SolarCity alone				=	Combined company			
Opeating Assets	36 444	Debt	3 942		Opeating Assets	5 957	Debt	2 857		Opeating Assets	42 401	Debt	6 799
							Minority Interests	981		Synergies	975	Minority Interests	981
		Equity	32 502				Equity	2 119		TOTAL:	43 376	Equity	35 596
		TOTAL:	36 444				TOTAL:	5 957				TOTAL:	43 376
		Number of shares (mn):	148				Number of shares (mn):	100				Exchange ratio (average):	0,1265
		Price per share:	219,61				Price per share:	21,19				Number of shares (mn):	160,7
												Price per share:	221,58

- **Offer = $28.03** = 0.1265 x $221.58
- **Premium = 32.3%** = $28.03 / $21.19 – 1
- **Premium (in $mn) = 684 =** ($ 28.03 – $ 21.19) x 100 **(70% of Synergies!)**

Value of Initial Offer on June 22, 2016 (Ex-Post)

Tesla alone

Opeating Assets	32 803	Debt	3 697
		Equity	29 106
		TOTAL:	32 803
		Number of shares (mn):	148
		Price per share:	196,66
		Variation from June, 21	-10,45%

+

SolarCity alone

Opeating Assets	6 156	Debt	2 987
		Minority Interests	981
		Equity	2 188
		TOTAL:	6 156
		Number of shares (mn):	100
		Price per share:	21,88
		Variation from June, 21	3,26%

=

Combined company

Opeating Assets	42 401	Debt	6 684
Synergies	(3 443)	Minority Interests	981
TOTAL:	38 959	Equity	31 294
		TOTAL:	38 959
		Exchange ratio (average):	0,1265
		Number of shares (mn):	160,7
		Price per share:	194,79
		Variation from June, 21	-12,09%

- **SCTY's debt went up… why? Where did this value come from?**
- Implied Synergies = Market Implied EV of 39.0 – Operating Assets (June 21) = **-3.4 bn.**
  - **The market initially didn't like it!**

Market reaction September 29, 2016 (Pre-write)

Tesla alone				+	SolarCity alone				=	Combined company			
Opeating Assets	38 698	Debt	3 949		Opeating Assets	6 673	Debt	3 022		Opeating Assets (*)	45 200	Debt	6 971
							Minority Interests	981		Synergies	171	Minority Interests	981
										TOTAL:	45 371		
		Equity	34 749				Equity	2 670				Equity	37 419
		TOTAL:	38 698				TOTAL:	6 673				TOTAL:	45 371
		Number of shares (mn):	148				Number of shares (mn):	100				Exchange ratio (average):	0,11
												Number of shares (mn):	159,0
		Price per share:	234,79				Price per share:	26,70				Price per share:	235,34
		Variation from June, 21	6,91%				Variation from June, 21	26,00%				Variation from June, 21	6,21%

(*) Adjusts for 6.6% move in NASDAQ

The market warmed to the idea!

Shareholders and Debtholders gains (Pre-write)

		% Change from 6/21	
	jun/21	jun/22	jul/29
SCTY Conv. Debt	$563m	20%	25%
SCTY Solar Bonds	$175m	11%	14%
SCTY Equity	$2,119	3%	26%
TSLA Equity	$32,500m	-10%	7%
TSLA Conv. Debt	$3,264	-8%	0%
NASDAQ		0%	7%
SCTY Bond Yield (Ex.11)	25%	15%	13%
TSLA Bond Yield (Ex.6)	7.4%	9.1%	9.2%

- **Clearly the merger benefits SolarCity's Shareholders and Debtholders.**
- Shareholders due to the **premium of the merger**.
- Debtholders due to the **"bailout".** The risk of SolarCity after the merger is reduced, because all the assets and cash flows of the new merged company will now be used to pay SolarCity debt.

Governance issues: Board independence

Name	Title	Tesla Board	SolarCity Board
Brad W. Buss	Former CFO SCTY	X	
Robyn M. Denholm	CFO/COO of Juniper Networks	X	
Ira Ehrenpreis	General Partner with Technology Partners	X	
John H.N. Fisher	Managing Director, Draper Fisher Jurvetson		X
Antonio J. Gracias	CEO, Chairman of Valor	X	X
Steve Jurvetson	Managing Director, Draper Fisher Jurvetson	X	
Donald R. Kendall Jr	Managing Director & CEO of Kenmont		X
Elon Musk	TSLA Chairman and CEO; SCTY Chairman	X	X
Kimbal Musk	CEO of Medium	X	
Nancy E. Pfund	Founder/Managing Partner of DBL Investors		X
Lyndon Rive	SCTY CEO and co-founder		X
Peter Rive	SCTY CTO and co-founder		X
JB Straubel	CTO TSLA		X

- UBS:

"There are 8 board members in SolarCity. Musk is conflicted (>20% ownership in both companies). It's also known that Lyndon & Peter Rive are first cousins of Musk. In addition, Straubel is TSLA CTO. This leaves only 4 independent members. Gracias is conflicted being on both boards, Fisher's firm (Draper Fisher Jurvetson) is also on both boards, and Pfund's firm was an early investor in TSLA. Arguably ***this leaves Donald Kendall Jr. as the lone member on SCTY's board without any apparent Tesla relationship...****"*

Governance issues: Elon Musk and its cousins' interests

- **Elon Musk's interests:**
  - Direct: 20% of each company;
  - Indirect: SpaceX owned $165 mn of SolarCity's SolarBonds.
- **Peter and Lyndon Rive:**
  - Shareholders of SolarCity;
  - Lyndon Rive owned $16 mn of SolarCity's convertible notes.

Common Shareholders and Institutional Investors

Tesla		SolarCity	
Stockholder	**Percentage**	**Stockholder**	**Percentage**
Elon Musk	26.5%	Elon Musk	18.4%
FMR LLC	10.2%	FMR LLC	8.2%
Baillie Gifford & Co.	8.2%	Baillie Giffor & Co.	6.7%
T. Rowe Price Associates, Inc.	6.0%	T. Rowe Price Associates, Inc.	5.0%
Company Insiders	1.2%	Company Insiders	0.7%
Total	**52%**	**Total**	**39.0%**
Total Institutional Ownership	**65.81%**	**Total Institutional Ownership**	**40.32%**
Total Institutional Voting Power Without Musk	**89.50%**	**Total Institutional Voting Power Without Musk**	**49.40%**

Common Shareholders and Institutional Investors (Cont.)

- Elon Musk was not the only shareholder of both companies; there were other shareholders who voted and who may have had conflicts of interest.

- Institutional investors could determine the outcome of the proposed merger. Musk's decision to refrain from voting shifts overwhelming power towards institutional investors and away from individual investors.

- Institutional investors own 40.32% and 65.81% of SolarCity and Tesla respectively. Accounting for Musk's non-voting block of shares, institutional investor ownership shifts to 49,4% and 89.5% of outstanding shares respectively. **Individual investors would have to vote nearly unanimously to block the merger.**

- While skepticism abounds the proposed Tesla-SolarCity merger, individual investors were practically powerless in the merger vote.

What Happened?

The Tesla-SolarCity Merger

- TSLA completed the acquisition of SolarCity Corporation SCTY on Nov 21. Shareholder approval for the merger was received the week before. Owners of nearly 85% of the outstanding Tesla shares voted in favor of the deal.

- In November 2016, Tesla acquired SolarCity for $2.6 billion in an all-stock deal, merging the largest U.S. residential solar installer into Tesla to advance Elon Musk's clean energy ecosystem vision.

- The deal faced scrutiny as a "bailout" of a cash-strapped firm, but was finalized after shareholder approval, with SolarCity rebranded as Tesla Solar.

- Despite the rocky start, the move integrated solar into Tesla's portfolio, allowing them to offer a combined energy storage (Powerwall) and generation package.

The “Bailout”

- Just weeks after the merger, SolarCity auditor Ernst & Young concluded the company lacked “sufficient cash to meet its obligations” as a stand-alone company. In other words, SolarCity was nearly insolvent when Tesla shareholders voted to pay $2.6 billion for the solar panel maker and take on its debts, rescuing it from probable bankruptcy.

- Tesla shareholders alleged the company’s acquisition of the solar installer amounted to a bailout, pushed through by Musk who sat on both company boards at the time. The shareholders also alleged that Musk controlled the board of Tesla, even though he appeared to recuse himself from some deal negotiations concerning SolarCity.

- Musk denied that he placed any pressure on the Tesla board to go through with the transaction.

- Shareholders sued Musk, claiming the acquisition was a bailout and arguing Musk pressured the board to save his investment, but a Delaware court ruled in 2022 that the deal was fair and not a bailout.